4(a)	($\omega = 2\pi / T$ and $T = 2.2$ s so) $\omega = 2\pi / 2.2 = 2.9$ rad s^{-1}	**A1**
4(b)(i)	$\omega^2 = g / R$	**C1**
	$R = 9.81 / 2.86^2$ $= 1.2$ m	**A1**
4(b)(ii)	$v_0 = \omega x_0$	**C1**
	$= 2.9 \times 3.0 \times 10^{-2}$ $= 0.087$ m s^{-1}	**A1**
4(c)	smooth wave starting at 3.0 cm when $t = 0$	**B1**
	positions of peaks and troughs show same period (or slightly longer)	**B1**
	each peak and trough at lower amplitude than the previous one	**B1**